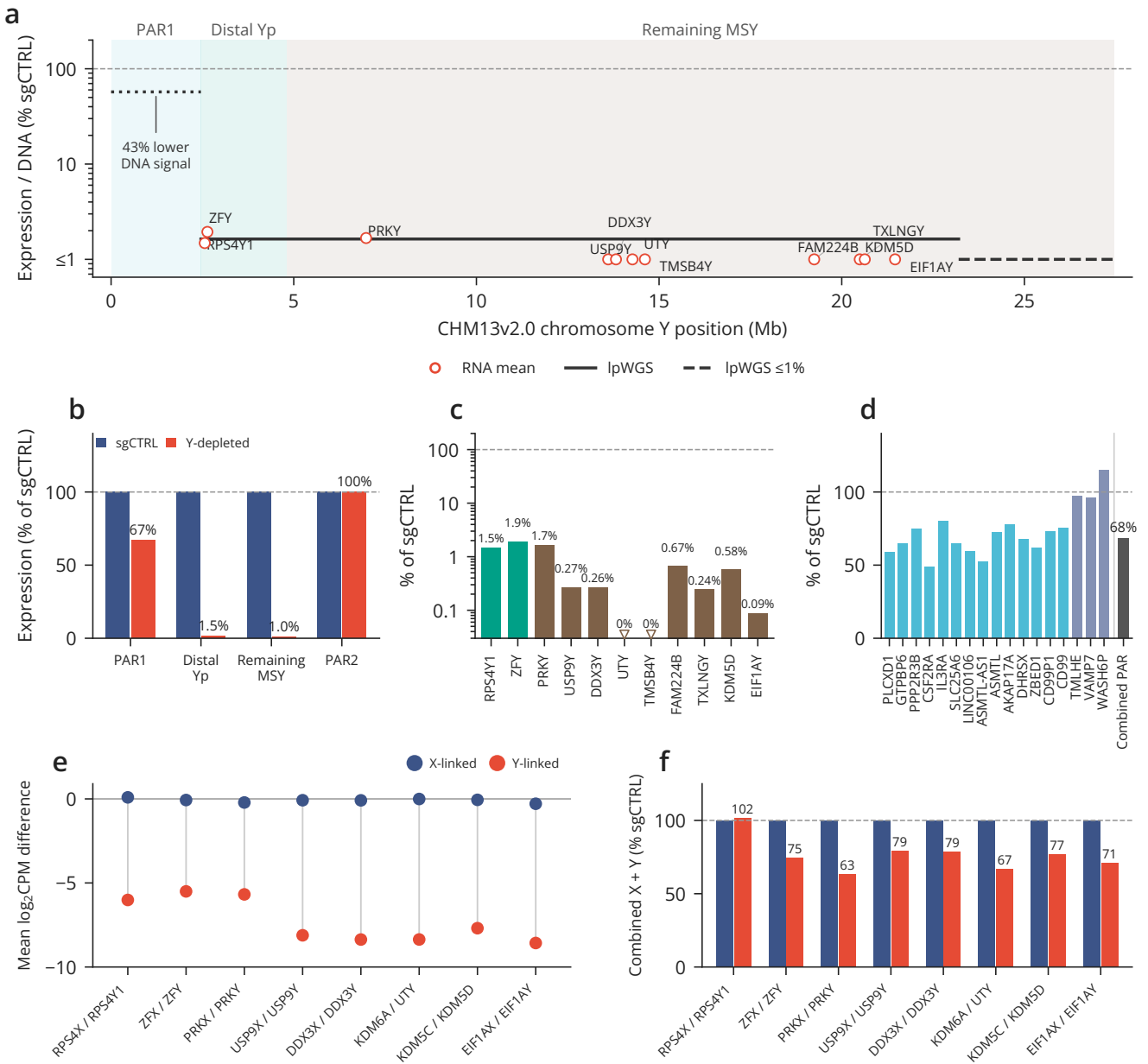
a
PAR1
Distal Yp
Remaining MSY
Expression / DNA (% sgCTRL)
100
10
≤1
43% lower DNA signal
ZFY
RPS4Y1
PRKY
DDX3Y
USP9Y
UTY
TMSB4Y
TXLNGY
FAM224B
KDM5D
EIF1AY
0
5
10
15
20
25
CHM13v2.0 chromosome Y position (Mb)
RNA mean
lpWGS
lpWGS ≤1%
b
sgCTRL
Y-depleted
Expression (% of sgCTRL)
100
50
0
67%
1.5%
1.0%
100%
PAR1
Distal Yp
Remaining MSY
PAR2
c
% of sgCTRL
100
10
1
0.1
1.5%
1.9%
1.7%
0.27%
0.26%
0%
0%
0.67%
0.24%
0.58%
0.09%
RPS4Y1
ZFY
PRKY
USP9Y
DDX3Y
UTY
TMSB4Y
FAM224B
TXLNGY
KDM5D
EIF1AY
d
% of sgCTRL
100
50
0
68%
PLCXD1
GTPBP6
PPP2R3B
CSF2RA
IL3RA
SLC25A6
LINC00106
ASMTL-AS1
ASMTL
AKAP17A
DHRSX
ZBED1
CD99P1
CD99
TMLHE
VAMP7
WASH6P
Combined PAR
e
X-linked
Y-linked
Mean log2CPM difference
0
−5
−10
RPS4X / RPS4Y1
ZFX / ZFY
PRKX / PRKY
USP9X / USP9Y
DDX3X / DDX3Y
KDM6A / UTY
KDM5C / KDM5D
EIF1AX / EIF1AY
f
Combined X + Y (% sgCTRL)
100
50
0
102
75
63
79
79
67
77
71
RPS4X / RPS4Y1
ZFX / ZFY
PRKX / PRKY
USP9X / USP9Y
DDX3X / DDX3Y
KDM6A / UTY
KDM5C / KDM5D
EIF1AX / EIF1AY